

More about games in extensive form with complete information

In this chapter we investigate some further aspects of games in extensive form with complete information.

First, we saw in Section 3.9 that when one converts a game in extensive form to normal form, new Nash equilibria can appear that do not correspond to the Nash equilibrium one finds by backward induction. The latter is special in that it gives a strategy one can sensibly use at any point in the game tree. In this chapter we give a name—subgame perfect Nash equilibria—and definition to these special Nash equilibria of a game in extensive form. The definition is general enough that it will help us deal with such games as the one in Figure 1.6, where the backward induction procedure we have used so far fails.

Second, we look at some infinite horizon games. Since infinite horizon games potentially last forever, they cannot be fully analyzed by backward induction. The notion of a subgame perfect Nash equilibrium will help us investigate these games.

Especially important are repeated games in which a prisoner's dilemma occurs over and over. In this context one can see why it might be advantageous to use the strictly dominated, cooperative strategy, if it induces cooperation from one's opponent in the future.

Infinite horizon games have more practical applications than one might think. For example, a game that is repeated an unknown number of times can often be thought of as an infinite horizon game.

Finally, we discuss two striking examples of games in extensive form with complete information, the Samaritan's Dilemma (Section 6.10) and the Rotten Kid Theorem (Section 6.11), that come from the work of Nobel Prize-winning economists. These two games were too complicated to include in Chapter 1, but in essence they are just applications of backward induction.

6.1 Subgame perfect Nash equilibria

Consider a game G in extensive form. A node c' is called a *successor* of a node c if there is a path in the game tree that starts at c and ends at c' .

Let h be a node that is not terminal and has no other nodes in its information set. Assume:

- If a node c is a successor of h , then every node in the information set of c is also a successor of h .

In this situation it makes sense to talk about the *subgame* H of G whose root is h . H consists of the node h and all its successors, connected by the same moves that connected them in G , and partitioned into the same information sets as in G . The players and the payoffs at the terminal vertices are also the same as in G .

If G is a game with complete information, then any nonterminal node of G is the root of a subgame of G .

Recall from Section 1.2 that in a game in extensive form, each node is at the end of a unique path from the root node. This is usually interpreted to mean that players remember the past. Thus you should think of a subgame as including the memory of what happened before the subgame began.

Let s_i be one of Player i 's strategies in the game G . Recall that s_i is just a plan for what move to make at every node labeled i in the game G . So of course s_i includes a plan for what move to make at every node labeled i in the subgame H . Thus s_i contains within it a strategy that Player i can use in the subgame H . We call this strategy the *restriction* of s_i to the subgame H , and label it s_{iH} .

Suppose the game G has n players, and $(s_1, \dots, s_n)$ is a Nash equilibrium for G . It is called a *subgame perfect Nash equilibrium* if, for every subgame H of G , $(s_{1H}, \dots, s_{nH})$ is a Nash equilibrium for H .

6.2 Big Monkey and Little Monkey 6

Recall the game of Big Monkey and Little Monkey from Section 1.5, with Big Monkey going first (Figure 6.1).

Recall that Big Monkey has two possible strategies in this game, and Little Monkey has four. When we find the payoffs for each choice of strategies, we get a game in normal form:

		Little Monkey			
		ww	wc	cw	cc
Big Monkey	w	(0, 0)	(0, 0)	(9, 1)	(9, 1)
	c	(4, 4)	(5, 3)	(4, 4)	(5, 3)

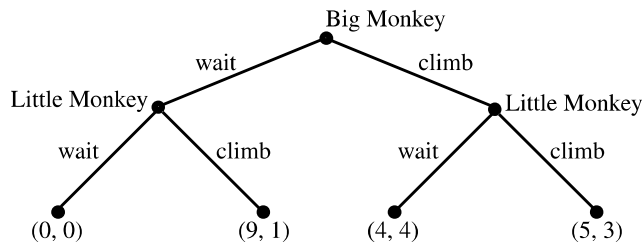


Figure 6.1. Big Monkey and Little Monkey.

We have seen (see Section 3.9) that there are three Nash equilibria in this game: (w, cw) , (c, ww) , and (w, cc) . The equilibrium (w, cw) , which is the one found by backward induction, is subgame perfect. The other two are not.

The equilibrium (c, ww) is not subgame perfect because, in the subgame that begins after Big Monkey waits, Little Monkey would wait. By switching to climb, Little Monkey would achieve a better payoff. Little Monkey's plan to wait if Big Monkey waits is what we called in Section 1.6 a *threat*: it would hurt Big Monkey, at the cost of hurting Little Monkey.

The equilibrium (w, cc) is not subgame perfect because, in the subgame that begins after Big Monkey climbs, Little Monkey would climb. By switching to wait, Little Monkey would achieve a better payoff. Little Monkey's plan to climb if Big Monkey climbs is what we called in Section 1.6 a *promise*: it would help Big Monkey, at the cost of hurting Little Monkey. As we saw there, in this particular game, the promise does not affect Big Monkey's behavior. Little Monkey is promising that if Big Monkey climbs, he will get a payoff of 5, rather than the payoff of 4 he would normally expect. Big Monkey ignores this promise because by waiting, he gets an even bigger payoff, namely, 9.

6.3 Subgame perfect equilibria and backward induction

When we find strategies in finite extensive-form games by backward induction, we are finding subgame perfect Nash equilibria. In fact, when we use backward induction, we are essentially considering every subgame in the entire game.

Strategies in a subgame perfect Nash equilibrium make sense no matter where in the game tree you use them. In contrast, at a Nash equilibrium that is not subgame perfect, at least one of the players is using a strategy that at some node tells her to make a move that would not be in her interest to make. For example, at a Nash equilibrium where a player is using a strategy that includes a threat, if the relevant node were reached, the strategy would

tell the player to make a move that would hurt her. The success of such a strategy depends on this node not being reached!

There are some finite games in extensive form for which backward induction does not work. Recall the game in Figure 6.2, which we discussed in Section 1.4.

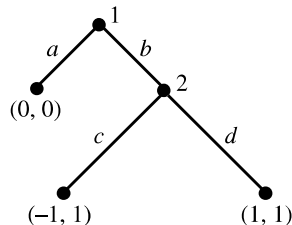


Figure 6.2. Failure of backward induction.

The problem with this game is that at the node where Player 2 chooses, both available moves give her a payoff of 1. Hence Player 1 does not know which move Player 2 will choose if that node is reached. However, Player 1 certainly wants to know which move Player 2 will choose before she decides between a and b !

In this game Player 1's strategy set is $S_1 = \{a, b\}$, and Player 2's strategy set is $S_2 = \{c, d\}$. In normal form, the game is just

		Player 2	
		c	d
Player 1	a	(0, 0)	(0, 0)
	b	(-1, 1)	(1, 1)

This game has two Nash equilibria, (a, c) and (b, d) . Both are subgame perfect.

There is a way to find all subgame perfect Nash equilibria in any finite game in extensive form with perfect information by a variant of backward induction. Do backward induction as usual. If at any point a player has *several* best choices, record each of them as a possible choice at that point, and *separately* continue the backward induction using each of them. Ultimately you will find all subgame perfect Nash equilibria.

For example, in the game we are presently considering, we begin the backward induction at the node where Player 2 is to choose, since it is the only node all of whose successors are terminal. Player 2 has *two* best choices, c and d . Continuing the backward induction using c , we find that Player 1 chooses a . Continuing the backward induction using d , we find that Player 1

chooses b . Thus the two strategy profiles we find are (a, c) and (b, d) . Both are subgame perfect Nash equilibria.

For a finite game in extensive form with complete information, this more general backward induction procedure never fails. Therefore *every finite game in extensive form with complete information has at least one subgame perfect Nash equilibrium*.

For a two-player zero-sum finite game in extensive form with complete information we know more (recall Section 5.5): any Nash equilibrium, and hence any subgame perfect Nash equilibrium, yields the same payoffs to the two players; and Player 1's strategy from one subgame perfect Nash equilibrium, played against Player 2's strategy from another subgame perfect Nash equilibrium, also yields those payoffs. Such strategies are "best" for the two players. This applies, for example, to chess: because of the rule that a game is a draw if a position is repeated three times, chess is a finite game. The game tree has about 10^{123} nodes. Since the tree is so large, best strategies for white (Player 1) and black (Player 2) are not known. In particular, it is not known whether the best strategies yield a win for white, a win for black, or a draw.

The notion of subgame perfect Nash equilibrium is especially valuable for infinite horizon games, for which backward induction cannot be used directly. These games are liable to have many Nash equilibria. Looking for the ones that are subgame perfect is a way of zeroing in on the (perhaps) most plausible ones. Sections 6.4–6.8 treat examples of infinite horizon games.

6.4 Duels and Truels

6.4.1 Duels. Two politicians, Newt and George, are competing for their party's presidential nomination. They take turns attacking each other. George is more skilled and better financed. Each time he attacks Newt, he has an 80% chance of knocking Newt out of the race. In contrast, each time Newt attacks George, he has a 50% chance of knocking George out of the race. The attacks continue until one candidate is driven from the race. If Newt attacks first, what is the probability that Newt is the survivor?

This problem is an example of a *duel*. It has several characteristics:

- (1) The players take turns attacking.
- (2) The duel could theoretically go on forever, if every attack fails to drive the opponent from the race.

In other duels, the players might attack simultaneously, so that a round could result in both players being driven from the race; this cannot happen when the players take turns. There are also duels that are not allowed to go on

forever; it could be known in advance that the duel will end after a certain number of rounds if both candidates survive that long.

Duels are not games. The players have no need to choose strategies. However, we treat this duel much like a game. We will represent it by a tree diagram (see Figure 6.3). The terminal vertices are those where a player has just been eliminated. At the terminal vertices, we assign a payoff of 1 to the surviving candidate and 0 to the other candidate. We want to determine each candidate's expected payoff, which is just the probability the he is the ultimate survivor.

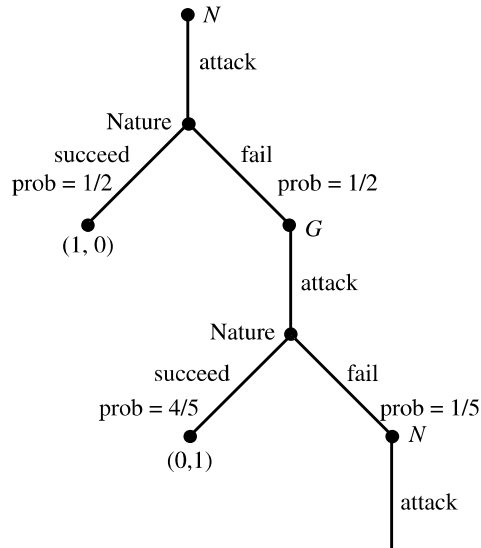


Figure 6.3. Tree diagram for the Duel. Player 1 is Newt (N), Player 2 is George (G).

Let π_1 denote Newt's expected payoff. If neither candidate is eliminated in the first two rounds, it will again be Newt's turn to attack. Since the situation will then be the same as it was at the start, Newt's expected payoff at that point will again be π_1 . This leads to the formula

$$\pi_1 = \frac{1}{2} \times 1 + \frac{1}{2} \times \left(\frac{4}{5} \times 0 + \frac{1}{5} \times \pi_1 \right).$$

Solving this equation for π_1 yields $\pi_1 = \frac{5}{9}$.

Thus Newt is the survivor $\frac{5}{9}$ of the time. Half the time he eliminates George with his first attack, and even if he fails, there is a chance he could eliminate George later.

6.4.2 Truels. A *truel* is like a duel except there are three contestants. Each time a contestant attacks, if he still has two surviving opponents, he must decide which opponent to attack. There are several types of truels:

- (1) The contestants can take turns attacking or can attack simultaneously.
- (2) The truel can be allowed to continue forever, or it can be required to end after a certain number of rounds.
- (3) When it is a contestant's turn to attack and both opponents remain, he may be required to attack an opponent, or he may be allowed to choose not to attack at all.

A truel in which the contestants take turns ends when there is only one surviving contestant. A truel in which the contestants attack simultaneously also ends if all three contestants are eliminated.

Unlike a duel, a truel is a game, since the players have several available strategies to choose from. As with a duel, when two contestants have been eliminated, we assign a payoff of 1 to the surviving contestant and 0 to the others. Payoffs also must be assigned to the case in which the game continues forever. The payoffs that make sense in this case depend on the context.

6.4.3 Analysis of a truel. Three politicians, Newt, George, and Ron, are competing for their party's presidential nomination. They take turns attacking each other. When it is a candidate's turn to attack, if both opponents remain in the race, he chooses one to attack, or he can choose to attack neither. If only one opponent remains in the race, the candidate attacks him. When Newt attacks a candidate, he has a 50% chance of knocking him out of the race. When George attacks a candidate, he has an 80% chance of knocking him out of the race. Ron, however, is a legendary supercandidate. When Ron attacks someone, Ron is sure to knock him out of the race. The attacks continue until only one candidate is left. If Newt attacks first, George second, and Ron third, what are each candidate's chances of winning?

We denote the initial state of the game by NGR . This means that it is Newt's turn; he will be followed by George, then Ron (if they are still in the race). Because of the candidates' order, the other possible states of the game with three players left are GRN and RNG . There are six possible two-candidate states (NG , NR , GN , GR , RN , and RG), and of course three possible one-candidate states, which correspond to terminal vertices in the game tree. However, in drawing game trees, we regard the two-person states as terminal, since for each two-person state, we can calculate the expected payoff of each player. We did this for the two-person state NG when we analyzed the duel between Newt and George. Here are the expected payoffs for all two-person states; Players 1, 2, and 3 are Newt, George, and Ron, respectively:

$$\begin{aligned}
 (\pi_1, \pi_2, \pi_3)(NG) &= \left(\frac{5}{9}, \frac{4}{9}, 0\right), & (\pi_1, \pi_2, \pi_3)(NR) &= \left(\frac{1}{2}, 0, \frac{1}{2}\right), \\
 (\pi_1, \pi_2, \pi_3)(GN) &= \left(\frac{1}{9}, \frac{8}{9}, 0\right), & (\pi_1, \pi_2, \pi_3)(GR) &= \left(0, \frac{4}{5}, \frac{1}{5}\right), \\
 (\pi_1, \pi_2, \pi_3)(RN) &= (0, 0, 1), & (\pi_1, \pi_2, \pi_3)(RG) &= (0, 0, 1).
 \end{aligned}$$

The payoffs for GN are calculated like those for NG . The payoffs for two-person states involving Ron are easier to compute, since if he gets a turn he will certainly eliminate his remaining opponent.

A player's strategy is a plan for what to do each time he encounters the one three-person state where it is his turn to attack. For example, Newt needs a plan for each time the state NGR ("his" state) is encountered. A priori his planned move could depend on all players' previous moves. However, we shall assume that each player plans the same move for every time his state is encountered. For example, if Newt plans to attack Ron on his initial turn, and if the state NGR is arrived at later in the game, Newt will attack Ron then too. It is reasonable to expect that there will be a subgame perfect Nash equilibrium with strategies of this type.

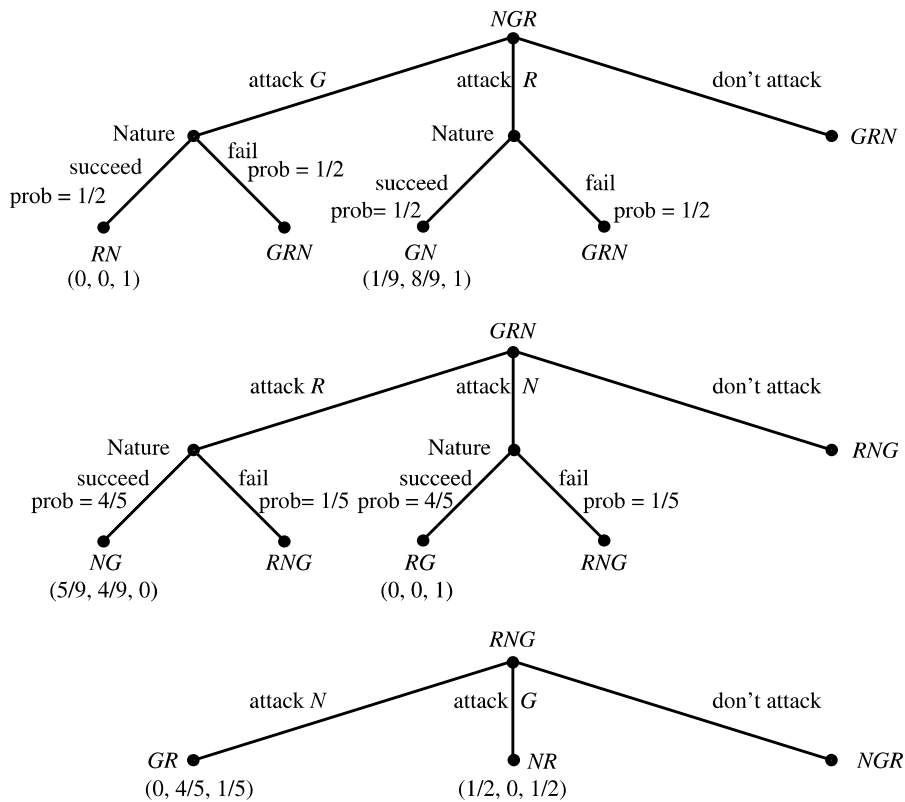


Figure 6.4. Partial game trees for the Truel. Notation is explained in the text.

Thus each player has only three strategies. For example, Newt can attack George (G), attack Ron (R), or not attack ($\emptyset$) whenever the state NGR is encountered.

Figure 6.4 shows partial game trees starting at the three-person states. From these trees we see:

- In the situation NGR , Newt's strategy R dominates his strategy G .
- In the situation GRN , George's strategy R dominates his strategy N .
- In the situation RNG , Ron's strategy G dominates his strategy N .

In other words, if you are going to attack an opponent, you should attack the stronger one. The reason is obvious: if you leave only the weaker of your opponents, you have a better chance of surviving. We therefore eliminate one strategy for each candidate, leaving two for each.

Now we can regard this game as one in normal form with three candidates and two strategies each. The payoffs are shown in Table 6.1. The strategy profile $(\emptyset, \emptyset, \emptyset)$ results in all candidates surviving forever; we need to assign payoffs to this case. In this problem, the most reasonable payoffs to assign seem to be $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, since if no candidate is ever eliminated, each presumably has probability $\frac{1}{3}$ of winning the nomination.

Table 6.1. Payoff matrices for the Truel

		Player 3 (Ron) attacks George	
		Player 2 (George)	
		R	$\emptyset$
Player 1 (Newt)	R	$(\frac{59}{180}, \frac{112}{180}, \frac{1}{20})$	$(\frac{11}{36}, \frac{4}{9}, \frac{1}{4})$
	$\emptyset$	$(\frac{49}{90}, \frac{16}{45}, \frac{1}{10})$	$(\frac{1}{2}, 0, \frac{1}{2})$
		Player 3 (Ron) does not attack	
		Player 2 (George)	
		R	$\emptyset$
Player 1 (Newt)	R	$(\frac{25}{81}, \frac{56}{81}, 0)$	$(\frac{1}{9}, \frac{8}{9}, 0)$
	$\emptyset$	$(\frac{5}{9}, \frac{4}{9}, 0)$	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

The table also shows best responses. There is one pure strategy Nash equilibrium. Newt, who starts the game, does not attack. The other two players each plan to attack their stronger opponent, who is not Newt. By thus lying low, Newt gets by far the best outcome of any player: with probability greater than half, he will be the lone survivor.

Some of the payoffs are easy to check. For example, if the strategy profile is $(R, \emptyset, G)$, Newt begins by attacking Ron. With probability $\frac{1}{2}$, Newt eliminates

Ron. The situation is now GN , which results in the payoffs $(\frac{1}{9}, \frac{8}{9}, 0)$. On the other hand, with probability $\frac{1}{2}$, Newt fails eliminate Ron. The situation is now GRN . Since George use the strategy $\emptyset$, he does not attack, and the situation becomes RNG . Since Ron uses the strategy G , he eliminates George, and the situation becomes NR . The payoffs from NR are $(\frac{1}{2}, 0, \frac{1}{2})$. Therefore

$$(\pi_1, \pi_2, \pi_3)(R, \emptyset, G) = \frac{1}{2}(\frac{1}{9}, \frac{8}{9}, 0) + \frac{1}{2}(\frac{1}{2}, 0, \frac{1}{2}) = (\frac{11}{36}, \frac{4}{9}, \frac{1}{4}).$$

Other payoffs can be checked by solving an equation. For example, if the strategy profile is $(R, R, \emptyset)$, Newt agains begins by attacking Ron. With probability $\frac{1}{2}$, Newt eliminates Ron, leading to the payoffs $(\frac{1}{9}, \frac{8}{9}, 0)$ as before. With probability $\frac{1}{2}$, Newt fails eliminate Ron, leading to the situation GRN . Since George now uses the strategy R , he attacks Ron. With probability $\frac{4}{5}$ he eliminates Ron, leading to the situation NG . This leads to the payoffs $(\frac{5}{9}, \frac{4}{9}, 0)$. With probability $\frac{1}{5}$, George fails to eliminate Ron. The situation becomes RNG . Since Ron uses the strategy $\emptyset$, he does not attack, and the situation becomes NGR : we are back where we started. Therefore

$$(\pi_1, \pi_2, \pi_3)(R, R, \emptyset) = \frac{1}{2}(\frac{1}{9}, \frac{8}{9}, 0) + \frac{1}{2}(\frac{4}{5}(\frac{5}{9}, \frac{4}{9}, 0) + \frac{1}{5}(\pi_1, \pi_2, \pi_3)).$$

The solution of this equation is $(\pi_1, \pi_2, \pi_3) = (\frac{25}{81}, \frac{56}{81}, 0)$.

We remark that we have found a Nash equilibrium under the assumption that each player is only allowed to use one of the three strategies we have described. Further analysis would be required to show that the strategy profile we have found remains a Nash equilibrium when other strategies are allowed and to show that it is subgame perfect.

6.5 The Rubinstein bargaining model

One dollar is to be split between two players. Player 1 goes first and offers to keep a fraction x_1 of the available money (one dollar). Of course, $0 \leq x_1 \leq 1$, and Player 2 would get the fraction $1 - x_1$ of the available money. If Player 2 accepts this proposal, the game is over, and the payoffs are x_1 to Player 1 and $1 - x_1$ to Player 2.

If Player 2 rejects the proposal, the money shrinks to δ dollars, $0 < \delta < 1$, and it becomes Player 2's turn to make an offer.

Player 2 offers a fraction y_1 of the available money (now δ dollars) to Player 1. Of course, $0 \leq y_1 \leq 1$, and Player 2 would get the fraction $1 - y_1$ of the available money. If Player 1 accepts this proposal, the game is over, and the payoffs are $y_1\delta$ to Player 1 and $(1 - y_1)\delta$ to Player 2.

If Player 1 rejects the proposal, the money shrinks to δ^2 , and it becomes Player 1's turn to make an offer.

Player 1 offers to keep a fraction x_2 of the available money (now δ^2 dollars) and give the fraction $1 - x_2$ to Player 2. ...Well, you probably get the idea. See Figure 6.5.

The payoff to each player is the money she gets. If no proposal is ever accepted, the payoff to each player is 0.

This game models a situation in which it is in everyone's interest to reach an agreement quickly. Think, for example, of labor negotiations during a strike: as the strike goes on, the workers lose pay, and the company loses production. However, you probably don't want to reach a quick agreement by offering everything to your opponent!

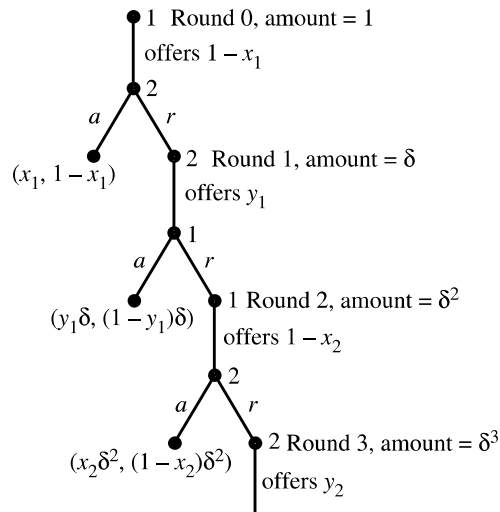


Figure 6.5. The Rubinstein bargaining model.

The numbering of the rounds of the game is shown in Figure 6.5.

A strategy for Player 1 consists of a plan for each round. For the even rounds, she must plan what offer to make. For the odd rounds, she must plan which offers to accept and which to reject. Of course, her plan can depend on what has happened up to that point.

Player 2's strategies are similar. For the even rounds, she must plan which offers to accept and which to reject. For the odd rounds, she must plan what offer to make.

Notice that at the start of any even round, Player 1 faces exactly the same situation that she faced at the start of the game, except that the available money is less. Similarly, at the start of any odd round, Player 2 faces exactly the same situation that she faced at the start of round 1, except that the available money is less.

We make two simplifying assumptions:

- (1) Suppose a player has a choice between accepting an offer (thus terminating the game) and rejecting the offer (thus extending the game), and suppose the payoff the player expects from extending the game equals the offer she was just made. Then she will accept the offer, thus terminating the game.
- (2) There is a subgame perfect Nash equilibrium with the following property: if it yields a payoff of x to Player 1, then in the subgame that starts at round 2, it yields a payoff of $x\delta^2$ to Player 1.

With these assumptions, in the subgame perfect equilibrium of assumption (2), if the game were to go to Round 2, the payoffs would be $(x\delta^2, (1-x)\delta^2)$. So the game tree gets pruned to the one shown in Figure 6.6.

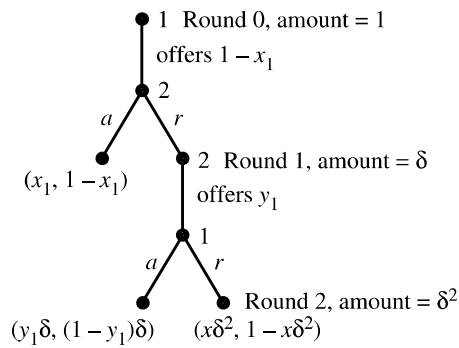


Figure 6.6. Pruned Rubinstein bargaining model.

Let us continue to investigate the subgame perfect equilibrium of assumption (2). Since it is subgame perfect, we should reason backward on the pruned game tree to find the players' remaining moves.

At round 1, Player 2 must make an offer. If she offers a fraction y_1^* of the available amount δ chosen so that $y_1^* \delta = x\delta^2$, Player 1 will be indifferent between accepting the offer and rejecting it. According to assumption 1, she will accept it. Player 2 will get to keep $\delta - y_1^* \delta$. If Player 2 offered more than y_1^* , Player 1 would accept, but Player 2 would get less than $\delta - y_1^* \delta$. If Player 2 offered less than y_1^* , Player 1 would not accept, and Player 2 would end up with $\delta^2 - x\delta^2$. This is less than $\delta - y_1^* \delta$:

$$\delta^2 - x\delta^2 = \delta^2 - y_1^* \delta < \delta - y_1^* \delta.$$

Thus Player 2's best move is to offer the fraction y_1^* of the available amount δ chosen so that $y_1^* \delta = x\delta^2$, that is, $y_1^* = x\delta$. Since this is her best move, it is

the move she makes at round 1 in our subgame perfect equilibrium. Player 1 accepts the offer.

At round 0, Player 1 must make an offer. If she offers a fraction $1 - x_1^*$ of the available amount 1 chosen so that $1 - x_1^* = (1 - y_1^*)\delta$, Player 2 will be indifferent between accepting the offer and rejecting it. According to assumption (1), she will accept it. Player one gets to keep x_1^* . Reasoning as before, we see that if Player 1 offers more or less than $1 - x_1^*$, she ends up with less than x_1^* . Thus Player 1's best move is to offer the fraction $1 - x_1^*$, Player 2 accepts, and Player 1's payoff is x_1^* .

From assumption (2), we conclude that $x_1^* = x$.

From the equations $1 - x = 1 - x_1^* = (1 - y_1^*)\delta$ and $y_1^* = x\delta$, we obtain

$$1 - x = (1 - y_1^*)\delta = (1 - x\delta)\delta = \delta - x\delta^2,$$

so $1 - \delta = x - x\delta^2$, and so

$$x = \frac{1 - \delta}{1 - \delta^2} = \frac{1}{1 + \delta}. \quad (6.1)$$

This is Player 1's payoff. Player 2's payoff is

$$1 - x = \frac{\delta}{1 + \delta}. \quad (6.2)$$

Player 1's payoff is higher. For δ close to 1 (i.e., when the money they are bargaining about does not shrink very fast), both payoffs are close to $\frac{1}{2}$.

6.6 Discount factor and repeated games

In Rubinstein's bargaining model, payoffs in the future are lower, because the amount of money to be divided decreases over time. However, there are good reasons to regard payoffs in the future as less valuable than payoffs today, even if they are not lower. One reason is that the future is always uncertain. A proverb says that a bird in the hand is worth two in the bush. Certainly a bird in the hand is worth more than one in the bush! A second reason is that money earns interest. If you can earn 4% annual interest, for example, a payoff of \$100 today is surely worth more than a payoff of \$100 in a year. If you had \$100 today, you could invest it and have \$104 in a year. That is why, if you win a big prize in a state lottery, you are offered either a certain amount now, or a larger amount over a period of years. A third reason is psychological. For most of us, the present is simply more important than the future. People differ, of course, in the relative importance they assign to the future.

In this section, we introduce the notion of discount factor, which formalizes the idea that payoffs in the future as less valuable than payoffs today. We then use the notion of discount factor to define repeated games, which will occupy us in the next two sections.

6.6.1 Discount factor. Suppose your boss proposes to you a salary of s dollars this year and t dollars next year. The utility to you of s dollars in the year you earn it is $u(s)$; recall the notion of a utility function introduced in Subsection 4.1.1. The total utility to you today of your boss's offer is $U(s, t) = u(s) + \delta u(t)$, where δ is a *discount factor* that is strictly between 0 and 1. The value of δ can reflect the uncertainty of the future, the interest rate you could earn, or the importance you attach to the future relative to the present.

Which would you prefer, a salary of s this year and s next year, or a salary of $s - a$ this year and $s + a$ next year? Assume $0 < a < s$, and, as we discussed in Subsection 4.1.1, assume $u' > 0$ (more is better) and $u'' < 0$ (the rate of increase of your utility decreases as your salary increases). Then

$$\begin{aligned} U(s, s) - U(s - a, s + a) &= u(s) + \delta u(s) - (u(s - a) + \delta u(s + a)) \\ &= u(s) - u(s - a) - \delta(u(s + a) - u(s)) \\ &= \int_{s-a}^s u'(t) dt - \delta \int_s^{s+a} u'(t) dt > 0. \end{aligned}$$

Hence you prefer s each year.

Do you see why the last line is positive? Part of the reason is that $u'(s)$ decreases as s increases, so $\int_{s-a}^s u'(t) dt > \int_s^{s+a} u'(t) dt$.

6.6.2 Definition of a repeated game. Let G be a game in normal form with players $1, \dots, n$, strategy sets $S_1, \dots, S_n$, and payoff functions $\pi_i(s_1, \dots, s_n)$.

We define a *repeated game* R with *stage game* G and *discount factor* δ , $0 < \delta < 1$, as follows. The stage game G is played at times $k = 0, 1, 2, \dots$, with *only pure strategies allowed*. A strategy for Player i is just a way of choosing which of his pure strategies s_i to use at each time k . His choice of which strategy to use at time k can depend on all the strategies used by all the players at times before k .

There will be a payoff to Player i from the stage game at each time $k = 0, 1, 2, \dots$. His payoff in the repeated game R is just

his payoff at time 0 + δ × his payoff at time 1 + δ^2 × his payoff at time 2 + $\dots$.

In other words, payoffs at time 1 are discounted by a factor δ , payoffs at time 2 by the factor δ^2 , and so forth.

A subgame of R is defined by taking the repeated game that starts at some time k . You should think of a subgame as including the memory of the strategies used by all players in the stage games at earlier times.

6.7 The Wine Merchant and the Connoisseur

A Wine Merchant sells good wine at \$5 a bottle. The cost of this wine to the Wine Merchant is \$4, so he makes \$1 profit on each bottle he sells. Instead of doing this, the Wine Merchant could try to sell terrible wine at \$4 a bottle. He can acquire terrible wine for essentially nothing.

Bernard is a regular customer and a wine connoisseur. He values the good wine at \$6 a bottle, so when he buys it for \$5, he feels he is ahead by \$1. If he ever tasted the terrible wine, he would value it at 0.

Bernard can either pay for his wine or steal it. If he steals the good wine, the Wine Merchant will complain to everyone, which will result in a loss of reputation to Bernard worth \$2. If Bernard steals the terrible wine, the Wine Merchant will not bother to complain.

We think of this situation as a two-player game in which the Wine Merchant has two strategies, sell good wine or sell bad wine. Bernard also has two strategies, pay for his wine or steal it.

Assuming Bernard wants one bottle of wine, we have the following game in normal form:

		Bernard	
		pay	steal
Wine Merchant	sell good	(1, 1)	(−4, 4)
	sell bad	(4, −4)	(0, 0)

The payoffs are explained as follows:

- (1, 1): The Wine Merchant makes \$1 profit when he sells good wine, Bernard gets a wine worth \$6 for \$5.
- (−4, 4): The Wine Merchant loses a bottle of wine that cost him \$4, Bernard gets a wine worth \$6 at the cost of a \$2 blow to his reputation.
- (4, −4): The Wine Merchant makes \$4 profit when he sells bad wine, Bernard has paid \$4 for wine that is worth nothing to him.
- (0, 0): The Wine Merchant loses a bottle of wine that cost him nothing, Bernard gets a wine worth nothing.

The Wine Merchant has a strictly dominant strategy: sell bad wine. Bernard also has a strictly dominant strategy: steal the wine. If both players use their nasty dominant strategies, each gets a payoff of 0. In contrast, if both players

use their nice, dominated strategies (The Wine Merchant sells good wine, Bernard pays for it), both get a payoff of 1. This game is a prisoner's dilemma.

To make this game easier to discuss, let's call each player's nice strategy c for "cooperate," and let's call each player's nasty strategy d for "defect." Now we have the following payoff matrix:

		Bernard	
		c	d
Wine merchant	c	(1, 1)	(-4, 4)
	d	(4, -4)	(0, 0)

Both players have the same strategy set, namely, $\{c, d\}$.

We will take this game to be the stage game in a repeated game R . Recall that a player's strategy in a repeated game is a way of choosing which of his strategies to use at each time k . The choice can depend on what all players have done at times before k .

We consider the *trigger strategy* in this repeated game, which we denote σ and which is defined as follows. Start by using c . Continue to use c as long as your opponent uses c . If your opponent ever uses d , use d at your next turn, and continue to use d forever.

In other words, the Wine Merchant starts by selling good wine, but if his customer steals it, he decides to minimize his losses by selling terrible wine in the future. Bernard starts by buying from the Wine Merchant, but if the Wine Merchant cheats him, he says to himself, "I'm not going to pay that cheater any more!"

Theorem 6.1. *If $\delta \geq \frac{3}{4}$, then (σ, σ) is a Nash equilibrium.*

Proof. If both players use σ , then both cooperate in every round. Therefore both receive a payoff of 1 in every round. Taking into account the discount factor δ , we have

$$\pi_1(\sigma, \sigma) = \pi_2(\sigma, \sigma) = 1 + \delta + \delta^2 + \cdots = \frac{1}{1 - \delta}.$$

Here we have used the formula for the sum of an infinite geometric series:

$$r + r\delta + r\delta^2 + \cdots = \frac{r}{1 - \delta} \quad \text{provided } |\delta| < 1.$$

Suppose Bernard switches to a different strategy σ' . (Because of the symmetry of the game, the argument would be exactly the same for the Wine Merchant.)

Case 1. Bernard still ends up cooperating in every round. Then the Wine Merchant, who is still using σ , will also cooperate in every round. The payoffs are unchanged.

Case 2. Bernard first defects in round k . Then the Wine Merchant will cooperate through round k , will defect in round $k+1$, and will defect in every round after that. Does using σ' improve Bernard's payoff?

The payoffs from the strategy profiles (σ, σ) and (σ, σ') are the same through round $k-1$, so let's just compare their payoffs to Bernard from round k on.

With (σ, σ) , Bernard's payoff from round k on is

$$\delta^k + \delta^{k+1} + \dots = \frac{\delta^k}{1-\delta}.$$

With (σ, σ') , Bernard's payoff in round k is 4: the payoff from stealing good wine. From round $k+1$ on, unfortunately, the Wine Merchant will defect (sell bad wine) in every round. Bernard's best response to this is to steal it, giving him a payoff of 0. Therefore, taking into account the discount factor, Bernard's payoff from round k on is at most

$$4\delta^k + 0(\delta^{k+1} + \delta^{k+2} + \dots) = 4\delta^k.$$

From round k on, Bernard's payoff from (σ, σ) is greater than or equal to his payoff from (σ, σ') provided

$$\frac{\delta^k}{1-\delta} \geq 4\delta^k \quad \text{or} \quad \delta \geq \frac{3}{4}.$$

□

Some people consider this way of analyzing repeated games to be game theory's most important contribution. It is said to show that cooperative behavior can establish itself in a society without the need for an external enforcer. In particular, it shows that cooperating in a prisoner's dilemma, as long as the other player does, can be rational when the game is to be repeated indefinitely. The lower bound for δ can be interpreted as meaning that that this conclusion holds when the players value the future highly enough.

Instead of using a discount factor, one can think of δ as the probability that the game will be repeated, given that it has just been played. The mathematics is exactly the same. The interpretation now is that if the probability of the game's being repeated is high enough, then it can be rational to use the cooperative strategy in a prisoner's dilemma.

Is (σ, σ) a subgame perfect Nash equilibrium? To answer this question, we must be sure we understand what σ tells us to do in every conceivable situation, including those in which σ has not been used correctly up to that point. For example, suppose your opponents defects, so you begin defecting, but in round j you make a mistake and accidentally cooperate. Should you go

back to defecting in round $j + 1$, or should you check whether your opponent perhaps cooperated in round j , and base your decision on that? We have not defined σ carefully enough to answer this question.

Let's redefine the trigger strategy σ as follows. Start by using c . In round j , if both you and your opponent have used c in all previous rounds, use c . Otherwise, use d .

You should be able to convince yourself that with this precise definition, (σ, σ) is a subgame perfect Nash equilibrium. Remember that a subgame is defined by starting the game at some time, with the memory of all that has happened previously. For any such subgame, you should convince yourself that if your opponent is using the trigger strategy σ as we have just redefined it, you can do no better than to use σ yourself.

6.8 The Folk Theorem

Are there other Nash equilibria in the repeated game of Wine Merchant and Connoisseur? The so-called Folk Theorem of Repeated Games says that there are many. (In mathematics, a “folk theorem” is a theorem that lots of people seem to know, but no one knows who first proved it.)

6.8.1 Payoff vectors. For a game in normal form with n players, let $\sigma = (\sigma_1, \dots, \sigma_n)$ be a mixed strategy profile. The *payoff vector* associated with σ is the n -tuple of numbers $(\pi_1(\sigma), \dots, \pi_n(\sigma))$. An n -tuple of numbers $(v_1, \dots, v_n)$ is a *possible payoff vector* for the game if it is the payoff vector associated with some strategy profile σ .

For example, consider a two-player game in normal form in which each player has two strategies:

		Player 2	
		t_1	t_2
Player 1	s_1	(a, b)	(c, d)
	s_2	(e, f)	(g, h)

If Player 1 uses the strategy $\sigma_1 = ps_1 + (1 - p)s_2$ with $0 \leq p \leq 1$, and Player 2 uses the strategy $\sigma_2 = qt_1 + (1 - q)t_2$ with $0 \leq q \leq 1$, then the payoff vector associated with the strategy profile (σ_1, σ_2) is

$$pq(a, b) + p(1 - q)(c, d) + (1 - p)q(e, f) + (1 - p)(1 - q)(g, h). \quad (6.3)$$

The set of all possible payoff vectors for the game can be thought of as a set of points in the plane. This set of points can be drawn as follows:

- (1) Draw the line segment from (a, b) to (c, d) . Think of it as parametrized by q , $0 \leq q \leq 1$, as follows: $r(q) = q(a, b) + (1 - q)(c, d)$.
- (2) Draw the line segment from (e, f) to (g, h) . Think of it as parametrized by q , $0 \leq q \leq 1$, as follows: $s(q) = q(e, f) + (1 - q)(g, h)$.
- (3) For each q , draw the line segment from $r(q)$ to $s(q)$. This set of points is parameterized by p , $0 \leq p \leq 1$, as follows:

$$pr(q) + (1 - p)s(q) = p(q(a, b) + (1 - q)(c, d)) + (1 - p)(q(e, f) + (1 - q)(g, h)).$$

Compare (6.3).

- (4) The union of all the lines you have drawn is the set of all possible payoff vectors for the game.

Figure 6.7 shows an example.

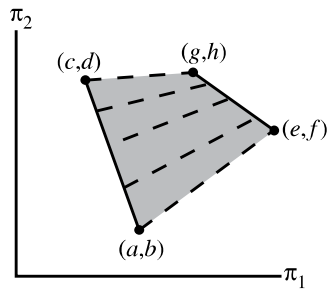


Figure 6.7. A set of possible payoff vectors. Typical line segments from $r(q)$ to $s(q)$ are dashed, and the set of all possible payoff vectors is gray.

6.8.2 Minimax payoffs. For a two-player game in normal form, Player 1's *minimax payoff* is the lowest payoff she ever gets when she makes her best response to one of Player 2's *pure* strategies. Player 1's minimax payoff is the lowest that Player 2 can force Player 1 to accept when using pure strategies. Player 2's minimax payoff is defined analogously.

To state the definition of minimax payoff symbolically, let s_i denote one of Player 1's pure strategies, and let t_j denote one of Player 2's pure strategies. Then Player 1's minimax payoff is $\min_{t_j} \max_{s_i} \pi_1(s_i, t_j)$. Player 2's minimax payoff is $\min_{s_i} \max_{t_j} \pi_2(s_i, t_j)$.

For example, in the game of Wine Merchant and Connoisseur (Section 6.7), if Bernard uses c , the Wine Merchant's best response is d , which gives him a payoff of 4. If Bernard uses d , the Wine Merchant's best response is again d , which gives him a payoff of 0. The minimum of 4 and 0 is 0, so the Wine Merchant's minimax payoff is 0. Bernard can force the Wine Merchant to accept a

payoff of 0 by using d , but he cannot force the Wine Merchant to accept any lower payoff. This fact was used in Bernard's trigger strategy to encourage the Wine Merchant to use c . If the Wine Merchant used d instead, Bernard would retaliate by using d , which would force the Wine Merchant to accept a payoff of 0. Since 0 is the Wine Merchant's minimax payoff, this is the worst punishment that Bernard can inflict.

6.8.3 The Folk Theorem. One version of the Folk Theorem says the following.

Theorem 6.2. *The Folk Theorem. Let G be a two-player game in normal form, let m_1 be Player 1's minimax payoff in G , and let m_2 be Player 2's minimax payoff in G . Let R be the repeated game with stage game G and discount factor δ . Then:*

- (1) *At a Nash equilibrium of R , Player 1's payoff in every round is at least m_1 , and Player 2's payoff in every round is at least m_2 .*
- (2) *Let (v_1, v_2) be a possible payoff vector for G with $v_1 > m_1$ and $v_2 > m_2$. Then there is a Nash equilibrium of R in which the average payoff vector is (v_1, v_2) , provided the discount factor δ is large enough.*

The average payoff vector is defined by taking the average payoff vector for the first k rounds, and then taking the limit as $k \rightarrow \infty$.

To show conclusion (1), we will just consider Player 1's payoff. Suppose we have arrived at round k of the repeated game. Player 2's strategy requires him to look at the history so far and choose a certain pure strategy t_j to use in round k . But then Player 1 can use the following strategy for round k : look at the history so far and choose the best response to t_j . This will give Player 1 a payoff of at least m_1 in round j .

We will not show (2) in general; we will only consider the case in which Players 1 and 2 each have two pure strategies. Let σ_1 and σ_2 be the strategies in G such that $\pi_1(\sigma_1, \sigma_2) = v_1$ and $\pi_2(\sigma_1, \sigma_2) = v_2$. Of course there are numbers p and q between 0 and 1 such that $\sigma_1 = ps_1 + (1-p)s_2$ and $\sigma_2 = qt_1 + (1-q)t_2$.

We only consider the case in which p and q are rational. We write these fractions as $p = \frac{k}{m}$ and $q = \frac{\ell}{n}$ with k, ℓ, m , and n integers.

We first describe a strategy $\tilde{\sigma}_2$ for Player 2 in the repeated game:

- First $q \cdot mn$ rounds: use t_1 . Note that $q \cdot mn = \frac{\ell}{n} \cdot mn = \ell m$, which is an integer, as it should be.
- Next $(1-q) \cdot mn = mn - \ell m$ rounds: use t_2 .

We have described Player 2's strategy for the first mn rounds. He does the same thing for the next mn rounds, and in fact forever.

We now describe a strategy $\tilde{\sigma}_1$ for Player 1 in the repeated game:

- First $p \cdot q \cdot mn$ rounds: use s_1 . Note that $p \cdot q \cdot mn = \frac{k}{m} \cdot \frac{\ell}{n} \cdot mn = k\ell$, which is an integer, as it should be.
- Next $(1 - p) \cdot q \cdot mn = \ell m - k\ell$ rounds: use s_2 .
- Next $p \cdot (1 - q) \cdot mn$ rounds: use s_1 . Note that $p \cdot (1 - q) \cdot mn = kn - k\ell$, which is an integer, as it should be.
- Next $(1 - p) \cdot (1 - q) \cdot mn = mn - kn - \ell m + k\ell$ rounds: use s_2 .

We have described Player 1's strategy for the first mn rounds. He does the same thing for the next mn rounds, and in fact forever.

Player 1's total payoff for the first mn rounds is

$$pqmn\pi_1(s_1, t_1) + (1 - p)qmn\pi_1(s_2, t_1) + p(1 - q)mn\pi_1(s_1, t_2) + (1 - p)(1 - q)mn\pi_1(s_2, t_2).$$

To get Player 1's average payoff for the first mn rounds, divide by mn :

$$\begin{aligned} & pq\pi_1(s_1, t_1) + (1 - p)q\pi_1(s_2, t_1) \\ & + p(1 - q)\pi_1(s_1, t_2) + (1 - p)(1 - q)\pi_1(s_2, t_2) \\ & = \pi_1(ps_1 + (1 - p)s_2, qt_1 + (1 - q)t_2) \\ & = \pi_1(\sigma_1, \sigma_2) \\ & = v_1. \end{aligned}$$

Similarly, Player 2's average payoff for the first mn rounds is $\pi_2(\sigma_1, \sigma_2) = v_2$, so the average payoff vector over the first mn rounds is (v_1, v_2) . This plus the repetitive character of $\tilde{\sigma}_1$ and $\tilde{\sigma}_2$ imply that the average payoff vector for the repeated game from the strategy profile $(\tilde{\sigma}_1, \tilde{\sigma}_2)$ is also (v_1, v_2) .

Now we define trigger strategies for Players 1 and 2 in the repeated game. Player 1: Start by using $\tilde{\sigma}_1$. If Player 2 ever deviates from his strategy $\tilde{\sigma}_2$, use the strategy s_i to which Player 2's best response gives him payoff $m_2 < v_2$, and continue to use it forever. Player 2: Start by using $\tilde{\sigma}_2$. If Player 1 ever deviates from his strategy $\tilde{\sigma}_1$, use the strategy t_j to which Player 1's best response gives him payoff $m_1 < v_1$, and continue to use it forever. The profile of these strategies is the desired Nash equilibrium of the repeated game for sufficiently large δ . (Without a sufficiently large discount factor, Player 1, for example, could at some stage use his best response to Player 2's action at that stage, instead of using the action prescribed by his strategy $\tilde{\sigma}_1$. This would give him a better payoff at that stage; with a small enough discount factor, he would also have a better payoff in the repeated game.)

Is the profile of these trigger strategies subgame perfect? Only if s_i and t_j are best responses to each other. If they are, then if we get to the point

where both players are using these strategies to punish the other, neither will be able to improve himself by changing on his own. This is what happens in prisoner's dilemma games, such as Wine Merchant and Connoisseur.

6.8.4 Wine Merchant and Connoisseur. For the game of Wine Merchant and Connoisseur, Figure 6.8 shows the possible payoff vectors (v_1, v_2) with $v_1 > 0$ and $v_2 > 0$ (since 0 is the minimax payoff for both players). According to the Folk Theorem, for any (v_1, v_2) in this set, there is a Nash equilibrium of the repeated game that gives payoffs (v_1, v_2) in every round, provided the discount factor δ is large enough.

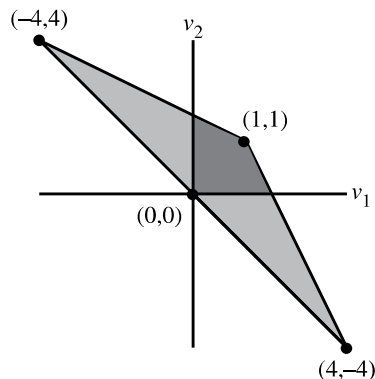


Figure 6.8. Possible payoff vectors for the game of Wine Merchant and Connoisseur (light gray). Those with $v_1 > 0$ and $v_2 > 0$ are in dark gray.

6.9 Maximum value of a function

Sections 6.10 and 6.11 discuss uses of backward induction by two Nobel prize-winning economists. To prepare for these examples, in this section we review some facts about the maximum value of a function.

Suppose f is a continuous function on an interval $a \leq x \leq b$. From calculus we know:

- (1) f attains a maximum value somewhere on the interval.
- (2) The maximum value of f occurs at a point where $f' = 0$, or at a point where f' does not exist, or at an endpoint of the interval.
- (3) If $f'(a) > 0$, the maximum does not occur at a .
- (4) If $f'(b) < 0$, the maximum does not occur at b .

Suppose that $f'' < 0$ everywhere in the interval $a \leq x \leq b$. Then we know a few additional things:

- (1) f attains its maximum value at *unique* point c in $[a, b]$.
- (2) Suppose $f'(x_0) > 0$ at some point $x_0 < b$. Then $x_0 < c$. See Figure 6.9.
- (3) Suppose $f'(x_1) < 0$ at some point $x_1 > a$. Then $c < x_1$.

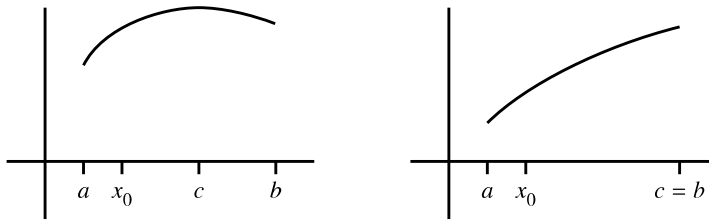


Figure 6.9. Two functions on $[a, b]$ with negative second derivative everywhere and positive first derivative at a point $x_0 < b$. Such functions always attain their maximum at a point c to the right of x_0 .

6.10 The Samaritan's Dilemma

There is someone you want to help should she need it. However, you are worried that the very fact that you are willing to help may lead her to do less for herself than she otherwise would. This is the Samaritan's Dilemma.

The Samaritan's Dilemma is an example of *moral hazard*. Moral hazard is “the prospect that a party insulated from risk may behave differently from the way it would behave if it were fully exposed to the risk.” There is a Wikipedia article on moral hazard: http://en.wikipedia.org/wiki/Moral_hazard.

Here is an example of the Samaritan's Dilemma analyzed by James Buchanan (Nobel Prize in Economics, 1986; Wikipedia article http://en.wikipedia.org/wiki/James_M._Buchanan).

A young woman plans to go to college next year. This year she is working and saving for college. If she needs additional help, her father will give her some of the money he earns this year.

Notation and assumptions regarding income and savings are as follows:

- (1) Father's income this year is $z > 0$, which is known. Of this he will give $0 \leq t \leq z$ to his daughter next year.
- (2) Daughter's income this year is $y > 0$, which is also known. Of this she saves $0 \leq s \leq y$ to spend on college next year.
- (3) Daughter chooses the amount s of her income to save for college. Father then observes s and chooses the amount t to give to his daughter.

The important point is (3): after Daughter is done saving, Father will choose an amount to give to her. Thus Daughter, who goes first in this game, can use

backward induction to figure out how much to save. In other words, she can take into account that different savings levels will result in different levels of support from Father.

The Utility functions (see Subsection 4.1.1) are

- (1) Daughter's utility function $\pi_1(s, t)$, which is her payoff in this game, is the sum of
 - (a) her first-year utility v_1 , a function of the amount she has to spend in the first year, which is $y - s$; and
 - (b) her second-year utility v_2 , a function of the amount she has to spend in the second year, which is $s + t$. Second-year utility is multiplied by a discount factor $\delta > 0$.

Thus we have

$$\pi_1(s, t) = v_1(y - s) + \delta v_2(s + t).$$

- (2) Father's utility function $\pi_2(s, t)$, which is his payoff in this game, is the sum of
 - (a) his personal utility u , a function of the amount he has to spend in the first year, which is $z - t$; and
 - (b) his daughter's utility π_1 , multiplied by a "coefficient of altruism" $\alpha > 0$.

Thus we have

$$\pi_2(s, t) = u(z - t) + \alpha \pi_1(s, t) = u(z - t) + \alpha(v_1(y - s) + \delta v_2(s + t)).$$

Notice that a component of Father's utility is Daughter's utility. The Samaritan's Dilemma arises when the welfare of someone else is important to us.

We assume (recall Subsection 4.1.1)

- (A1) The functions v_1 , v_2 , and u have positive first derivatives and negative second derivatives.

Let's first gather some facts that we will use in our analysis.

- (1) Formulas we will need for partial derivatives. They are calculated using the chain rule of first-semester calculus:

$$\begin{aligned}\frac{\partial \pi_1}{\partial s}(s, t) &= -v_1'(y - s) + \delta v_2'(s + t), \\ \frac{\partial \pi_2}{\partial t}(s, t) &= -u'(z - t) + \alpha \delta v_2'(s + t).\end{aligned}$$

(2) Formulas we will need for second partial derivatives:

$$\begin{aligned}\frac{\partial^2 \pi_1}{\partial s^2}(s, t) &= v_1''(\gamma - s) + \delta v_2''(s + t), \\ \frac{\partial^2 \pi_2}{\partial s \partial t}(s, t) &= \alpha \delta v_2''(s + t), \\ \frac{\partial^2 \pi_2}{\partial t^2}(s, t) &= u''(z - t) + \alpha \delta v_2''(s + t).\end{aligned}$$

All three of these are negative everywhere.

To find a savings level for Daughter using backward induction, we must first figure out, for each possible savings level s of Daughter, the contribution t that Father will make. To do this, we maximize Father's utility $\pi_2(s, t)$ with s fixed and $0 \leq t \leq z$ (recall that z is Father's income). Let's keep things from getting too complicated by arranging that for s fixed, $\pi_2(s, t)$ will attain its maximum at some t strictly between 0 and z . In other words, no matter how much Daughter saves, Father will give her some of his income but not all. This is guaranteed to happen if $(\partial \pi_2 / \partial t)(s, 0) > 0$ and $(\partial \pi_2 / \partial t)(s, z) < 0$. The first condition prevents Father from giving Daughter nothing. The second prevents him from giving Daughter everything.

For $0 \leq s \leq \gamma$, we have

$$\frac{\partial \pi_2}{\partial t}(s, 0) = -u'(z) + \alpha \delta v_2'(s) \geq -u'(z) + \alpha \delta v_2'(\gamma)$$

and

$$\frac{\partial \pi_2}{\partial t}(s, z) = -u'(0) + \alpha \delta v_2'(s + z) \leq -u'(0) + \alpha \delta v_2'(z).$$

We therefore make two more assumptions:

(A2) $\alpha \delta v_2'(\gamma) > u'(z)$. This assumption is reasonable. We expect Daughter's income γ to be much less than Father's income z . Since, as discussed in Subsection 4.1.1, each dollar of added income is less important when income is higher, we expect $v_2'(\gamma)$ to be much greater than $u'(z)$. If the product $\alpha \delta$ is not too small (meaning that Father cares quite a bit about Daughter, and Daughter cares quite a bit about the future), we get our assumption.

(A3) $u'(0) > \alpha \delta v_2'(z)$. This assumption is reasonable because $u'(0)$ should be large and $v_2'(z)$ should be small.

With these assumptions, we have

$$\frac{\partial \pi_2}{\partial t}(s, 0) > 0 \quad \text{and} \quad \frac{\partial \pi_2}{\partial t}(s, z) < 0 \quad \text{for all } 0 \leq s \leq \gamma.$$

Since $\partial^2 \pi_2 / \partial t^2$ is always negative, from Section 6.9, there is a single value of t where $\pi_2(s, t)$, s fixed, attains its maximum value; moreover, $0 < t < z$, so

$(\partial\pi_2/\partial t)(s, t) = 0$ at this value of t . We denote this value of t by $t = b(s)$. This is Father's best-response strategy, the amount Father will give to Daughter if the amount Daughter saves is s .

Daughter now chooses her saving level $s = s^*$ to maximize the function $\pi_1(s, b(s))$, which we shall denote $V(s)$:

$$V(s) = \pi_1(s, b(s)) = v_1(y - s) + \delta v_2(s + b(s)).$$

Father then contributes $t^* = b(s^*)$.

Here is the punchline: suppose it turns out that $0 < s^* < y$; that is, Daughter saves some of her income but not all. (This is the usual case.) Then, had Father simply committed himself in advance to providing t^* in support to his daughter no matter how much she saved, Daughter would have chosen a savings rate $s^\#$ greater than s^* . *Both* Daughter and Father would have ended up with higher utility.

To see this, note that we have

$$\frac{\partial\pi_1}{\partial s}(s^*, t^*) = -v'_1(y - s^*) + \delta v'_2(s^* + t^*). \quad (6.4)$$

Suppose that we can show that this expression is positive. Then, since $(\partial^2\pi_1/\partial s^2)(s, t^*)$ is always negative, we have that $\pi_1(s, t^*)$ is maximum at a value $s = s^\#$ greater than s^* . (See Section 6.9.)

We of course have $\pi_1(s^\#, t^*) > \pi_1(s^*, t^*)$, so Daughter's utility is higher. Since Daughter's utility is higher, we see from the formula for π_2 that $\pi_2(s^\#, t^*) > \pi_2(s^*, t^*)$, so Father's utility is also higher.

However, it is not obvious that (6.4) is positive. To see that it is, we proceed as follows.

Step 1. To maximize $V(s)$, we calculate

$$V'(s) = -v'_1(y - s) + \delta v'_2(s + b(s))(1 + b'(s)).$$

Step 2. If $V(s)$ is maximum at $s = s^*$ with $0 < s^* < y$, we must have $V'(s^*) = 0$, that is,

$$0 = -v'_1(y - s^*) + \delta v'_2(s^* + t^*)(1 + b'(s^*)). \quad (6.5)$$

Step 3. Subtracting (6.5) from (6.4), we obtain

$$\frac{\partial\pi_1}{\partial s}(s^*, t^*) = -\delta v'_2(s^* + t^*)b'(s^*). \quad (6.6)$$

Step 4. We expect that $b'(s) < 0$; this simply says that if Daughter saves more, Father will contribute less. To check this, note that

$$\frac{\partial\pi_2}{\partial t}(s, b(s)) = 0 \quad \text{for all } s.$$

Differentiating both sides of this equation with respect to s , we get

$$\frac{\partial^2 \pi_2}{\partial s \partial t}(s, b(s)) + \frac{\partial^2 \pi_2}{\partial t^2}(s, b(s))b'(s) = 0.$$

Since $\partial^2 \pi_2 / \partial s \partial t$ and $\partial^2 \pi_2 / \partial t^2$ are always negative, we must have $b'(s) < 0$.

Step 5. From (6.6), since v'_2 is always positive and $b'(s)$ is always negative, we see that $(\partial \pi_1 / \partial s)(s^*, t^*)$ is positive.

This problem has implications for government social policy. It suggests that social programs be made available to everyone rather than on an if-needed basis.

Let's look more closely at this conclusion.

When Father promises Daughter a certain fixed amount of help, one can imagine two possible effects: (i) now that she knows she will get this help, Daughter will save less; (ii) now that more saving will not result in less contribution from Father (remember, $b'(s) < 0$), Daughter will save more. All we have shown is that *if the promised contribution is t^** , it is actually (ii) that will occur. Too great a promised contribution might result in (i) instead.

In addition, our conclusion required that the coefficient of altruism α not be too small. That makes sense for a father and daughter. Whether it is correct for rich people (who do most of the paying for social programs) and poor people (who get most of the benefits) is less certain.

6.11 The Rotten Kid Theorem

A rotten son manages a family business. The amount of effort the son puts into the business affects both his income and his mother's. The son, being rotten, cares only about his own income, not his mother's. To make matters worse, Mother dearly loves her son. If the son's income is low, Mother will give part of her own income to her son so that he will not suffer. In this situation, can the son be expected to do what is best for the family?

We shall give the analysis of Gary Becker (Nobel Prize in Economics, 1992; Wikipedia article http://en.wikipedia.org/wiki/Gary_Becker).

We denote the son's annual income by y and the mother's by z . The amount of effort that the son devotes to the family business is denoted by a . His choice of a will affect both his income and his mother's, so we regard both y and z as functions of a : $y = y(a)$ and $z = z(a)$.

After Mother observes a , and hence observes her own income $z(a)$ and her son's income $y(a)$, she chooses an amount t , $0 \leq t \leq z(a)$, to give to her son.

The mother and son have personal utility functions u and v , respectively (see Subsection 4.1.1). Each is a function of the amount they have to spend.

The son chooses his effort a to maximize his own utility v , without regard for his mother's utility u . Mother, however, chooses the amount t to transfer to her son to maximize $u(z - t) + \alpha v(y + t)$, where α is her coefficient of altruism. Thus the payoff functions for this game are

$$\begin{aligned}\pi_1(a, t) &= v(y(a) + t), \\ \pi_2(a, t) &= u(z(a) - t) + \alpha v(y(a) + t).\end{aligned}$$

Since the son chooses first, he can use backward induction to decide how much effort to put into the family business. In other words, he can take into account that even if he doesn't put in much effort, and so doesn't produce much income for either himself or his mother, his mother will help him out.

We make the following assumptions:

- As in Subsection 4.1.1, the functions u and v have positive first derivatives and negative second derivatives.
- The son's level of effort is chosen from an interval $I = [a_1, a_2]$.
- For all a in I , $\alpha v'(y(a)) > u'(z(a))$. This assumption expresses two ideas: (a) Mother dearly loves her son, so α is not small; and (b) no matter how little or how much the son works, Mother's income $z(a)$ is much larger than son's income $y(a)$. (Recall that the derivative of a utility function gets smaller as the income gets larger.) This makes sense if the income generated by the family business is small compared to Mother's overall income
- For all a in I , $u'(0) > \alpha v'(y(a) + z(a))$. This assumption is reasonable, because $u'(0)$ should be large and $v'(y(a) + z(a))$ should be small.
- Let $T(a) = y(a) + z(a)$ denote total family income. Then $T'(a) = 0$ at a unique point $a^\#$, $a_1 < a^\# < a_2$, and $T(a)$ attains its maximum value at this point. This assumption expresses the idea that if the son works too hard, he will do more harm than good. As they say in the software industry, if you stay at work too late, you're just adding bugs.

To find the son's level of effort using backward induction, we must first maximize $\pi_2(a, t)$ with a fixed and $0 \leq t \leq z(a)$. We calculate

$$\begin{aligned}\frac{\partial \pi_2}{\partial t}(a, t) &= -u'(z(a) - t) + \alpha v'(y(a) + t), \\ \frac{\partial \pi_2}{\partial t}(a, 0) &= -u'(z(a)) + \alpha v'(y(a)) > 0, \\ \frac{\partial \pi_2}{\partial t}(a, z(a)) &= -u'(0) + \alpha v'(y(a) + z(a)) < 0, \\ \frac{\partial^2 \pi_2}{\partial t^2}(a, t) &= u''(z(a) - t) + \alpha v''(y(a) + t) < 0.\end{aligned}$$

Then there is a single value of t where $\pi_2(a, t)$, a fixed, attains its maximum; moreover, $0 < t < z(a)$, so $(\partial\pi_2/\partial t)(a, t) = 0$. (See Section 6.9.) We denote this value of t by $t = b(a)$. This is Mother's strategy, the amount Mother will give to her son if his level of effort in the family business is a .

The son now chooses his level of effort $a = a^*$ to maximize the function $\pi_1(a, b(a))$, which we shall denote $V(a)$:

$$V(a) = \pi_1(a, b(a)) = v(y(a) + b(a)).$$

Mother then contributes $t^* = b(a^*)$.

So what? Here is Becker's point.

Suppose $a_1 < a^* < a_2$ (the usual case). Then $V'(a^*) = 0$, that is,

$$v'(y(a^*) + b(a^*))(y'(a^*) + b'(a^*)) = 0.$$

Since v' is positive everywhere, we have

$$y'(a^*) + b'(a^*) = 0. \quad (6.7)$$

Now $(\partial\pi_2/\partial t)(a, t) = 0$ when $t = b(a)$, so $-u'(z(a) - b(a)) + \alpha v'(y(a) + b(a)) = 0$ for all a . Differentiating this equation with respect to a , we find that, for all a ,

$$-u''(z(a) - b(a))(z'(a) - b'(a)) + \alpha v''(y(a) + b(a))(y'(a) + b'(a)) = 0.$$

In particular, for $a = a^*$,

$$\begin{aligned} & -u''(z(a^*) - b(a^*))(z'(a^*) - b'(a^*)) \\ & + \alpha v''(y(a^*) + b(a^*))(y'(a^*) + b'(a^*)) = 0. \end{aligned}$$

This equation and (6.7) imply that

$$z'(a^*) - b'(a^*) = 0.$$

Adding this equation to (6.7), we obtain

$$y'(a^*) + z'(a^*) = 0.$$

Therefore $T'(a^*) = 0$. But then, by our last assumption, $a^* = a^\#$, the level of effort that maximizes total family income.

Thus, if the son had not been rotten, and instead had been trying to maximize total family income $y(a) + z(a)$, he would have chosen the same level of effort a^* .